Provability relations among modal axioms in systems

The Open Logic Project contributors

openlogic:normal-modal-logic/axioms-systems:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: item (6) carries the sub-label prop:S5facts-KT-D inside the same statement; this single card covers the whole proposition (same-span family consolidation), and the derivations use the dual axioms of def:duals with the rewriting marks 'with ... for p' delivered verbatim

Editorial note. The label prop:S5facts names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Dual axioms. The dual axioms T_Diamond (p implies Diamond p), B_Diamond (Diamond Box p implies p), 4_Diamond (Diamond Diamond p implies Diamond p) and 5_Diamond (Diamond Box p implies Box p) are available in the systems containing their box counterparts, and D is its own dual.. Several derivations take an instance of a dual axiom or rewrite a Box/Diamond form through one (the source marks these as 'with ... for p').
- The RK rule and propositional closure. Every normal modal logic is closed under RK and under propositional reasoning (PL), and contains its defining axiom instances.. Lines marked RK apply the rule to a previously derived implication; final lines combine derived implications by PL.

Proposition 0.1. *The following provability results obtain:*

1. **KT5** $\vdash$ B*;*
2. **KT5** $\vdash$ 4*;*

3. **KDB4** $\vdash$ T*;*

4. **KB4** $\vdash$ 5*;*

5. **KB5** $\vdash$ 4*;*

6. **KT** $\vdash$ D.

Proof. We exhibit proofs for each.

1. **KT5** $\vdash$ B:

   1. **KT5** $\vdash \Diamond A \to \Box\Diamond A$ 5
   2. **KT5** $\vdash A \to \Diamond A$ $\mathrm{T}_\Diamond$
   3. **KT5** $\vdash A \to \Box\Diamond A$ PL, 2, 1.

2. **KT5** $\vdash$ 4:

   1. **KT5** $\vdash \Diamond\Box A \to \Box\Diamond\Box A$ 5 with $\Box A$ for p
   2. **KT5** $\vdash \Box A \to \Diamond\Box A$ $\mathrm{T}_\Diamond$ with $\Box A$ for p
   3. **KT5** $\vdash \Box A \to \Box\Diamond\Box A$ PL, 2, 1
   4. **KT5** $\vdash \Diamond\Box A \to \Box A$ $5_\Diamond$
   5. **KT5** $\vdash \Box\Diamond\Box A \to \Box\Box A$ RK, 4
   6. **KT5** $\vdash \Box A \to \Box\Box A$ PL, 3, 5.

3. **KDB4** $\vdash$ T:

   1. **KDB4** $\vdash \Diamond\Box A \to A$ $\mathrm{B}_\Diamond$
   2. **KDB4** $\vdash \Box\Box A \to \Diamond\Box A$ D with $\Box A$ for p
   3. **KDB4** $\vdash \Box\Box A \to A$ PL, 1, 2
   4. **KDB4** $\vdash \Box A \to \Box\Box A$ 4
   5. **KDB4** $\vdash \Box A \to A$ PL, 4, 3.

4. **KB4** $\vdash$ 5:

   1. **KB4** $\vdash \Diamond A \to \Box\Diamond\Diamond A$ B with $\Diamond A$ for p
   2. **KB4** $\vdash \Diamond\Diamond A \to \Diamond A$ $4_\Diamond$
   3. **KB4** $\vdash \Box\Diamond\Diamond A \to \Box\Diamond A$ RK, 2
   4. **KB4** $\vdash \Diamond A \to \Box\Diamond A$ PL, 1, 3.

5. **KB5** $\vdash$ 4:

   1. **KB5** $\vdash \Box A \to \Box\Diamond\Box A$ B with $\Box A$ for p
   2. **KB5** $\vdash \Diamond\Box A \to \Box A$ $5_\Diamond$
   3. **KB5** $\vdash \Box\Diamond\Box A \to \Box\Box A$ RK, 2
   4. **KB5** $\vdash \Box A \to \Box\Box A$ PL, 1, 3.

6. **KT** $\vdash$ D:

   1. **KT** $\vdash \Box A \to A$ T
   2. **KT** $\vdash A \to \Diamond A$ $\mathrm{T}_\Diamond$
   3. **KT** $\vdash \Box A \to \Diamond A$ PL, 1, 2

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